

BSDS Exercise Set 5

Detailed step-by-step L^AT_EX solutions with theory blocks

Question 1

Full question. Consider a random walk on the following graph:

$$\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}$$

- (a) Write the transition matrix.
- (b) If the walker starts at vertex 1, i.e. $\mathbb{P}(X_0 = 1) = 1$, what is the probability distribution after 2 steps, i.e. find $\mathbb{P}(X_2 = i)$ for $i = 1, 2, 3, 4$?
- (c) Is this chain doubly stochastic? Explain.

Quoted theory from the book. For a simple random walk on an undirected graph, the walker moves uniformly to one of the neighboring vertices. In a finite irreducible Markov chain, a stationary distribution exists, and for an undirected regular graph the uniform distribution is stationary.

Solution.

Step 1: Identify the edges of the graph.

The picture is the square graph with edges

$$1 \leftrightarrow 2, \quad 1 \leftrightarrow 3, \quad 2 \leftrightarrow 4, \quad 3 \leftrightarrow 4.$$

Each vertex has exactly two neighbors.

Step 2: Write the transition matrix.

Since the walk is uniform among the neighbors of the current vertex, each nonzero transition probability is $1/2$. Therefore

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.$$

Step 3: Compute the distribution after two steps from vertex 1.

Starting from 1, the chain must first move to 2 or 3:

$$\mathbb{P}(X_1 = 2 \mid X_0 = 1) = \frac{1}{2}, \quad \mathbb{P}(X_1 = 3 \mid X_0 = 1) = \frac{1}{2}.$$

From either 2 or 3, the chain moves to 1 or 4, each with probability $1/2$. Hence

$$\mathbb{P}(X_2 = 1 \mid X_0 = 1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2},$$

$$\mathbb{P}(X_2 = 4 \mid X_0 = 1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2},$$

and there is no way to be at 2 or 3 after exactly two steps:

$$\mathbb{P}(X_2 = 2 \mid X_0 = 1) = 0, \quad \mathbb{P}(X_2 = 3 \mid X_0 = 1) = 0.$$

So the distribution of X_2 is

$$\left(\mathbb{P}(X_2 = 1), \mathbb{P}(X_2 = 2), \mathbb{P}(X_2 = 3), \mathbb{P}(X_2 = 4) \right) = \left(\frac{1}{2}, 0, 0, \frac{1}{2} \right).$$

Step 4: Check whether the chain is doubly stochastic.

A transition matrix is doubly stochastic if each column also sums to 1. Here,

$$\text{column 1 : } 0 + \frac{1}{2} + \frac{1}{2} + 0 = 1, \quad \text{column 2 : } \frac{1}{2} + 0 + 0 + \frac{1}{2} = 1,$$

$$\text{column 3 : } \frac{1}{2} + 0 + 0 + \frac{1}{2} = 1, \quad \text{column 4 : } 0 + \frac{1}{2} + \frac{1}{2} + 0 = 1.$$

So the matrix is doubly stochastic.

Final answer.

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}, \quad \mathbb{P}(X_2 = \cdot \mid X_0 = 1) = \left(\frac{1}{2}, 0, 0, \frac{1}{2} \right),$$

and the chain is doubly stochastic.

Question 2

Full question. For any Markov chain $\{X_n : n \geq 0\}$ with transition matrix P , show that $\{Y_n := X_{nk} : n \geq 0\}$ is also a Markov chain for any $k \geq 2$. Find its transition matrix. Show by an example that the chain $\{Y_n : n \geq 0\}$ may not be irreducible even if the original chain is irreducible.

Quoted theory from the book. The Markov property says that the future depends on the present state only. If we look only every k steps, the new process still has the Markov property, and the k -step transition probabilities are the entries of P^k .

Solution.

Step 1: Write the conditional probability for the sampled chain.

Fix states $i_0, i_1, \dots, i_n, j$. Then

$$\mathbb{P}(Y_{n+1} = j \mid Y_n = i_n, \dots, Y_0 = i_0) = \mathbb{P}(X_{(n+1)k} = j \mid X_{nk} = i_n, \dots, X_0 = i_0).$$

Step 2: Apply the Markov property to the original chain.

Because $\{X_n\}$ is Markov, the future after time nk depends only on $X_{nk} = i_n$. Therefore

$$\mathbb{P}(X_{(n+1)k} = j \mid X_{nk} = i_n, \dots, X_0 = i_0) = \mathbb{P}(X_{(n+1)k} = j \mid X_{nk} = i_n).$$

This depends only on the current sampled state i_n , so $\{Y_n\}$ is itself a Markov chain.

Step 3: Identify the transition matrix.

By definition,

$$\mathbb{P}(Y_{n+1} = j \mid Y_n = i) = \mathbb{P}(X_{(n+1)k} = j \mid X_{nk} = i) = p_{ij}^{(k)},$$

where $p_{ij}^{(k)}$ is the k -step transition probability of the original chain. Hence the transition matrix of $\{Y_n\}$ is

$$\boxed{P^k}.$$

Step 4: Give an irreducible chain whose sampled chain is not irreducible.

Take the two-state chain

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This chain is irreducible: from each state one reaches the other in one step.

Now sample every 2 steps. Then

$$P^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

So the sampled chain never moves from its starting state. Its states are absorbing, and the chain is not irreducible.

Final answer.

$Y_n = X_{nk}$ is Markov, with transition matrix P^k .

An irreducible original chain can produce a reducible sampled chain, as shown by the two-state alternating chain above.

Question 3

Full question. Consider a Markov chain on the state space $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & \frac{2}{3} & 0 \end{pmatrix}.$$

- (a) Find a recurrence relation between $p_{11}^{(n)}$ and hence find $p_{11}^{(n)}$.
- (b) Compute $f_{11}^{(n)}$. Show that the state is positive recurrent. Compute $\mu_1 = \sum_{n=0}^{\infty} n f_{11}^{(n)}$.

Quoted theory from the book. For a recurrent state i , the first return probabilities $f_{ii}^{(n)}$ and the return probabilities $p_{ii}^{(n)}$ satisfy the renewal equation

$$p_{ii}^{(n)} = \sum_{m=1}^n f_{ii}^{(m)} p_{ii}^{(n-m)},$$

with $p_{ii}^{(0)} = 1$. Also, if the state is positive recurrent, then its mean return time is finite.

Solution.

Step 0: A note on the matrix.

The scan is slightly garbled in the last row. To keep the chain stochastic and consistent with the intended layout, I use the stochastic completion

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{1}{4} \\ \frac{2}{3} & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}.$$

This is the version used below.

Step 1: Compute $p_{11}^{(n)}$.

The key observation is that states 1 and 4 have the same outgoing probabilities. Therefore, after two steps the chain is again in the set $\{1, 4\}$, and the law on $\{1, 4\}$ depends only on the fact that we started from 1.

From state 1,

$$\mathbb{P}(X_2 = 1 \mid X_0 = 1) = \frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{2}{3} = \frac{17}{24}, \quad \mathbb{P}(X_2 = 4 \mid X_0 = 1) = 1 - \frac{17}{24} = \frac{7}{24}.$$

Since the rows for states 1 and 4 are identical, the same two-step law applies whenever the chain is at 1 or 4. Therefore, after every two more steps, the probability of being at 1 is unchanged. Hence

$$p_{11}^{(0)} = 1, \quad p_{11}^{(1)} = 0, \quad p_{11}^{(n+2)} = p_{11}^{(n)} \text{ for } n \geq 2.$$

So

$$p_{11}^{(n)} = \begin{cases} 1, & n = 0, \\ 0, & n \text{ odd}, \\ \frac{17}{24}, & n \geq 2 \text{ even}. \end{cases}$$

Step 2: Find the first return probabilities $f_{11}^{(n)}$.

Let

$$P_{11}(s) = \sum_{n \geq 0} p_{11}^{(n)} s^n, \quad F_{11}(s) = \sum_{n \geq 1} f_{11}^{(n)} s^n.$$

The renewal identity gives

$$P_{11}(s) = \frac{1}{1 - F_{11}(s)}, \quad \text{so} \quad F_{11}(s) = 1 - \frac{1}{P_{11}(s)}.$$

Since

$$P_{11}(s) = 1 + \frac{17}{24} (s^2 + s^4 + s^6 + \cdots) = 1 + \frac{17}{24} \frac{s^2}{1 - s^2},$$

we get

$$F_{11}(s) = 1 - \frac{1}{1 + \frac{17}{24} \frac{s^2}{1 - s^2}} = \frac{17s^2}{24 - 7s^2}.$$

Expanding this as a power series,

$$F_{11}(s) = \frac{17}{24} s^2 \sum_{m \geq 0} \left(\frac{7}{24} s^2 \right)^m.$$

Therefore

$$f_{11}^{(2m+1)} = 0, \quad f_{11}^{(2m)} = \frac{17}{24} \left(\frac{7}{24} \right)^{m-1} \quad (m \geq 1).$$

Step 3: Show that the state is positive recurrent.

The mean return time is

$$\mu_1 = \sum_{n=1}^{\infty} n f_{11}^{(n)} = \sum_{m=1}^{\infty} 2m \cdot \frac{17}{24} \left(\frac{7}{24} \right)^{m-1}.$$

Using

$$\sum_{m=1}^{\infty} m r^{m-1} = \frac{1}{(1-r)^2}, \quad |r| < 1,$$

with $r = \frac{7}{24}$, we obtain

$$\mu_1 = 2 \cdot \frac{17}{24} \cdot \frac{1}{\left(1 - \frac{7}{24}\right)^2} = 2 \cdot \frac{17}{24} \cdot \frac{24^2}{17^2} = \frac{48}{17}.$$

This is finite, so state 1 is positive recurrent.

Final answer.

$$p_{11}^{(n)} = \begin{cases} 1, & n = 0, \\ 0, & n \text{ odd}, \\ \frac{17}{24}, & n \geq 2 \text{ even}, \end{cases} \quad f_{11}^{(2m)} = \frac{17}{24} \left(\frac{7}{24} \right)^{m-1}, \quad f_{11}^{(2m+1)} = 0,$$

and

$$\boxed{\mu_1 = \frac{48}{17}}.$$

Question 4

Full question. Consider the random walk on a graph $G = (V, E)$. Let $i, j \in V$.

(a) For $i_1, i_2, \dots, i_{n-1} \in V$, show that

$$\deg(i) \mathbb{P}(X_1 = i_1, X_2 = i_2, \dots, X_{n-1} = i_{n-1}, X_n = j \mid X_0 = i) = \deg(j) \mathbb{P}(X_1 = i_{n-1}, X_2 = i_{n-2}, \dots, X_n = i \mid X_0 = j)$$

(b) For any $n \geq 1$, show that

$$\deg(i) p_{ij}^{(n)} = \deg(j) p_{ji}^{(n)}.$$

Quoted theory from the book. For a simple random walk on an undirected graph, each step chooses a neighbor uniformly. The degree of a vertex counts the number of neighbors, and path probabilities are products of the corresponding reciprocal degrees.

Solution.

Step 1: Write the probability of a specific path.

For the path

$$i \rightarrow i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_{n-1} \rightarrow j,$$

the probability is

$$\frac{1}{\deg(i)} \cdot \frac{1}{\deg(i_1)} \cdots \frac{1}{\deg(i_{n-1})}.$$

Therefore,

$$\deg(i) \mathbb{P}(X_1 = i_1, \dots, X_n = j \mid X_0 = i) = \frac{1}{\deg(i_1) \cdots \deg(i_{n-1})}.$$

Step 2: Write the reversed path probability.

The reversed path is

$$j \rightarrow i_{n-1} \rightarrow i_{n-2} \rightarrow \dots \rightarrow i_1 \rightarrow i.$$

Its probability is

$$\frac{1}{\deg(j)} \cdot \frac{1}{\deg(i_{n-1})} \cdots \frac{1}{\deg(i_1)}.$$

So

$$\deg(j) \mathbb{P}(X_1 = i_{n-1}, \dots, X_n = i \mid X_0 = j) = \frac{1}{\deg(i_1) \cdots \deg(i_{n-1})}.$$

Step 3: Compare the two sides.

The right-hand sides are identical, so the required identity in part (a) follows immediately.

Step 4: Sum over all intermediate vertices.

Now sum the identity in part (a) over all possible choices of $i_1, \dots, i_{n-1} \in V$. The left side becomes $\deg(i) p_{ij}^{(n)}$ and the right side becomes $\deg(j) p_{ji}^{(n)}$. Hence

$$\deg(i) p_{ij}^{(n)} = \deg(j) p_{ji}^{(n)}.$$

Final answer. The path-reversal argument proves both parts.

Question 5

Full question. For any $i, j \in S$, show that

$$\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^* \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$

Quoted theory from the book. The hitting event $\{X_n = j \text{ for some } n \geq 1\}$ is the union of the disjoint events $\{T_j = n\}$. Also, if an event is contained in a union of events, its probability is at most the sum of the probabilities of those events.

Solution.

Step 1: Interpret f_{ij}^* .

The quantity f_{ij}^* is the probability that, starting from i , the chain ever hits j .

Step 2: Prove the lower bound.

For each fixed $n \geq 1$,

$$\{X_n = j\} \subseteq \{\text{the chain ever hits } j\}.$$

Therefore

$$p_{ij}^{(n)} \leq f_{ij}^*.$$

Taking the supremum over n gives

$$\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^*.$$

Step 3: Prove the upper bound.

If the chain ever hits j , then it must hit j for the first time at some finite time $n \geq 1$. Hence

$$\{\text{ever hit } j\} = \bigcup_{n \geq 1} \{T_j = n\},$$

where T_j is the first hitting time of j . These events are disjoint, so

$$f_{ij}^* = \sum_{n \geq 1} \mathbb{P}(T_j = n \mid X_0 = i).$$

Now $\{T_j = n\} \subseteq \{X_n = j\}$, so

$$\mathbb{P}(T_j = n \mid X_0 = i) \leq p_{ij}^{(n)}.$$

Summing over n yields

$$f_{ij}^* \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$

Final answer.

$$\sup_{n \geq 1} p_{ij}^{(n)} \leq f_{ij}^* \leq \sum_{n=1}^{\infty} p_{ij}^{(n)}.$$

Question 6

Full question. Suppose that $i \sim j$. Define the event E_{ij} that the chain visits j before returning to i starting from i . More precisely, for $n \geq 1$, set

$$g_{ij}^{(n)} = \mathbb{P}(X_n = j, X_t \neq i \text{ for } 1 \leq t < n \mid X_0 = i).$$

Show that

$$\mathbb{P}(E_{ij} \mid X_0 = i) = \sum_{n=1}^{\infty} g_{ij}^{(n)} > 0.$$

Quoted theory from the book. When a Markov chain is started from a state i , the event of first hitting a state j before returning to i can be decomposed into disjoint first-passage events over the possible hitting times.

Solution.

Step 1: Decompose the event E_{ij} .

The event E_{ij} means that, starting from i , the chain reaches j before it comes back to i . This can happen at some first time $n \geq 1$. The event that this first happens at time n is exactly

$$\{X_n = j, X_t \neq i \text{ for } 1 \leq t < n\}.$$

These events are disjoint for different n .

Step 2: Sum over all possible first-passage times.

Therefore,

$$\mathbb{P}(E_{ij} \mid X_0 = i) = \sum_{n=1}^{\infty} g_{ij}^{(n)}.$$

Step 3: Prove positivity.

Because $i \sim j$ and $i \neq j$, there exists a path

$$i = x_0, x_1, \dots, x_m = j$$

with positive transition probabilities along each edge. Choose such a path with no repeated visits to i after time 0; for instance, a shortest path in the directed graph of positive transitions works. Then the probability of following that path is a positive contribution to $g_{ij}^{(m)}$, so at least one $g_{ij}^{(n)}$ is positive. Hence the sum is strictly positive.

Final answer.

$$\mathbb{P}(E_{ij} \mid X_0 = i) = \sum_{n=1}^{\infty} g_{ij}^{(n)} > 0.$$

Question 7

Full question. Suppose $\{\varepsilon_i : i \geq 1\}$ be iid random variables taking values in $\mathbb{Z}$ such that $\mathbb{E}(\varepsilon_1) \neq 0$. Let

$$S_0 = 0, \quad S_n = \sum_{i=1}^n \varepsilon_i, \quad n \geq 1.$$

Show that the chain $\{S_n : n \geq 0\}$ must be transient.

Quoted theory from the book. The hint points to the strong law of large numbers: if $\mathbb{E}(\varepsilon_1)$ exists, then $S_n/n \rightarrow \mathbb{E}(\varepsilon_1)$ almost surely.

Solution.

Step 1: Apply the strong law of large numbers.

Let $\mu = \mathbb{E}(\varepsilon_1) \neq 0$. By the strong law,

$$\frac{S_n}{n} \rightarrow \mu \quad \text{almost surely.}$$

Step 2: Deduce that S_n drifts to infinity in one direction.

If $\mu > 0$, then $S_n \rightarrow +\infty$ almost surely. If $\mu < 0$, then $S_n \rightarrow -\infty$ almost surely. In either case, for any fixed integer a , the sequence $\{S_n\}$ can hit a only finitely many times almost surely.

Step 3: Translate this to transience.

A state is recurrent only if it is visited infinitely often with probability 1 when started from that state. Here, every state is visited only finitely many times almost surely, so the chain cannot be recurrent.

Final answer.

$\{S_n\}$ is transient.

Question 8

Full question. Consider the Markov Chain on $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Find all the stationary distributions.

Quoted theory from the book. A stationary distribution π satisfies $\pi = \pi P$. If a chain has a closed communicating class and all other states are transient, then every stationary distribution is supported on the closed class.

Solution.

Step 1: Identify the closed class.

States 1 and 2 communicate with each other, and from either 1 or 2 the chain never moves to 3 or 4. So $\{1, 2\}$ is a closed communicating class.

States 3 and 4 can move into $\{1, 2\}$, but once the chain enters $\{1, 2\}$, it never returns to 3 or 4. Therefore 3 and 4 are transient.

Step 2: Restrict the stationary distribution to the closed class.

Any stationary distribution must assign zero mass to transient states. So write

$$\pi = (\pi_1, \pi_2, 0, 0).$$

On the class $\{1, 2\}$ the reduced transition matrix is

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{2}{5} & \frac{3}{5} \end{pmatrix}.$$

Step 3: Solve the balance equations.

The equation $\pi = \pi P$ becomes

$$\pi_1 = \frac{1}{2}\pi_1 + \frac{2}{5}\pi_2, \quad \pi_1 + \pi_2 = 1.$$

From the first equation,

$$\frac{1}{2}\pi_1 = \frac{2}{5}\pi_2 \implies \pi_1 = \frac{4}{5}\pi_2.$$

Substitute into $\pi_1 + \pi_2 = 1$:

$$\frac{4}{5}\pi_2 + \pi_2 = 1 \implies \frac{9}{5}\pi_2 = 1 \implies \pi_2 = \frac{5}{9}.$$

Hence

$$\pi_1 = \frac{4}{9}.$$

Final answer.

$$\pi = \left(\frac{4}{9}, \frac{5}{9}, 0, 0 \right)$$

is the unique stationary distribution.

Question 9

Full question. Consider the Markov Chain on $S = \{1, 2, 3, 4\}$ with transition matrix

$$P = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{5} & \frac{2}{5} & \frac{1}{5} & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}.$$

Find all the stationary distributions.

Quoted theory from the book. A finite irreducible Markov chain has a unique stationary distribution. It is found by solving the balance equations $\pi = \pi P$ together with $\sum_i \pi_i = 1$.

Solution.

Step 1: Check irreducibility.

From state 1 we can go to 4, from 4 to 3, from 3 to 2, and from 2 back to 1. So all states communicate. Hence the chain is irreducible.

Step 2: Write the balance equations.

Let

$$\pi = (\pi_1, \pi_2, \pi_3, \pi_4).$$

Then $\pi = \pi P$ gives

$$\begin{aligned} \pi_1 &= \frac{1}{3}\pi_1 + \frac{2}{5}\pi_2 + \frac{1}{4}\pi_4, \\ \pi_2 &= \frac{1}{3}\pi_1 + \frac{2}{5}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \\ \pi_3 &= \frac{1}{5}\pi_2 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \\ \pi_4 &= \frac{1}{3}\pi_1 + \frac{1}{3}\pi_3 + \frac{1}{4}\pi_4, \end{aligned}$$

with

$$\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1.$$

Step 3: Solve the system.

Solving the linear equations gives

$$\pi_1 = \frac{33}{118}, \quad \pi_2 = \frac{20}{59}, \quad \pi_3 = \frac{21}{118}, \quad \pi_4 = \frac{12}{59}.$$

It is sometimes cleaner to write this as

$$\pi = \frac{1}{118}(33, 40, 21, 24).$$

Final answer.

$$\pi = \left(\frac{33}{118}, \frac{20}{59}, \frac{21}{118}, \frac{12}{59} \right).$$

Because the chain is finite and irreducible, this stationary distribution is unique.

Question 10

Full question. Consider two independent simple symmetric random walks X_n, Y_n . Consider the joint random walk restricted to $X_n \geq 0, Y_n \geq 0, X_n + Y_n \leq 10$. In other words, if any of the random walks try to escape the region, it is kept at the same place. Can you find the stationary distribution?

Quoted theory from the book. For a finite Markov chain, if a transition matrix is symmetric (or more generally satisfies detailed balance with the uniform law on a closed class), then the uniform distribution on that class is stationary.

Solution.

Step 1: Describe the state space.

The state space is

$$S = \{(x, y) \in \mathbb{Z}_{\geq 0}^2 : x + y \leq 10\}.$$

Its size is

$$|S| = 11 + 10 + \cdots + 1 = 66.$$

Step 2: Understand the parity structure.

At each step the pair moves by changing both coordinates by ± 1 unless the move would leave the triangle, in which case the chain stays put. Therefore the parity of $x + y$ is preserved. So S splits into two closed communicating classes:

$$E = \{(x, y) \in S : x + y \text{ is even}\}, \quad O = \{(x, y) \in S : x + y \text{ is odd}\}.$$

Counting points by diagonal sums gives

$$|E| = 1 + 3 + 5 + 7 + 9 + 11 = 36, \quad |O| = 2 + 4 + 6 + 8 + 10 = 30.$$

Step 3: Use symmetry.

The walk is symmetric inside each closed class, so the uniform distribution on each class is stationary:

$$u_E(z) = \frac{1}{36} \mathbf{1}_{\{z \in E\}}, \quad u_O(z) = \frac{1}{30} \mathbf{1}_{\{z \in O\}}.$$

Step 4: Describe all stationary distributions.

Because the chain has two closed communicating classes, every stationary distribution is a

convex combination of the stationary distributions on those classes:

$$\pi_\alpha = \alpha u_E + (1 - \alpha) u_O, \quad 0 \leq \alpha \leq 1.$$

In particular, the uniform distribution on all 66 states is stationary if we choose $\alpha = 36/66$.

Final answer.

$$\pi_\alpha = \alpha \frac{1}{36} \mathbf{1}_{\{z \in E\}} + (1 - \alpha) \frac{1}{30} \mathbf{1}_{\{z \in O\}}, \quad 0 \leq \alpha \leq 1.$$

Question 11

Full question. Consider an irreducible aperiodic Markov chain.

- (a) For any pair of states $i, j \in S$, show that there exists $M = M(i, j)$ such that $p_{ij}^{(n)} > 0$ for all $n \geq M$.
- (b) If the chain has N states, show that there exists $M = M(N)$ such that $p_{ij}^{(n)} > 0$ for all states i, j and $n \geq M$.
- (c) For Exercise 1, explicitly come up with a value M such that $p_{ij}^{(n)} > 0$ for any $i, j \in S$ and $n \geq M$.

Quoted theory from the book. In a finite irreducible aperiodic Markov chain, the return-time set of each state has greatest common divisor 1, so sufficiently large times can be realized by concatenating cycles. This implies that for large n , transition probabilities become strictly positive.

Solution.

Step 1: Use irreducibility.

If $i \sim j$, there is at least one path from i to j with positive probability. Let its length be r , so $p_{ij}^{(r)} > 0$.

Step 2: Use aperiodicity.

Aperiodicity means that for each state, the set of return times has gcd 1. In a finite chain this implies that, beyond some threshold, every large enough time can be written as a nonnegative combination of allowed cycle lengths. Therefore from i to j we can append enough cycles to the fixed positive path of length r and obtain positive probability for all large n .

So there exists $M(i, j)$ such that

$$p_{ij}^{(n)} > 0 \quad \text{for all } n \geq M(i, j).$$

Step 3: Make the bound uniform over all pairs.

If the chain has finitely many states, there are only finitely many pairs (i, j) . Let

$$M(N) = \max_{i, j \in S} M(i, j).$$

Then

$$p_{ij}^{(n)} > 0 \quad \text{for all } i, j \in S, \quad n \geq M(N).$$

Step 4: Address Exercise 1.

For Exercise 1, the graph is the 4-cycle

$$1 \leftrightarrow 2 \leftrightarrow 4 \leftrightarrow 3 \leftrightarrow 1.$$

This chain is irreducible but it is *periodic of period 2* because it is bipartite. Hence there is no single M such that $p_{ij}^{(n)} > 0$ for all i, j and all $n \geq M$.

So part (c) is impossible for Exercise 1 as stated.

Final answer. Parts (a) and (b) are true for finite irreducible aperiodic chains. For Exercise 1, no such M exists because the chain has period 2.

Question 12

Full question. Suppose $\{X_n : n \geq 0\}$ is a Markov chain. For some integer $K \geq 1$ and a state i_0 , suppose that

$$\inf\{p_{i,i_0}^{(K)} : i \in S\} > 0.$$

Show that

$$\mathbb{P}(\tau_{i_0} > n \mid X_0 = i) \leq C_1 e^{-C_2 n}$$

for some $C_1, C_2 > 0$ and for any $i \in S$, where

$$\tau_k = \inf\{n \geq 1 : X_n = k\}.$$

Quoted theory from the book. If from every state there is a uniformly positive chance to hit a fixed state after K steps, then repeated application of the Markov property gives geometric decay of the probability of avoiding that state for many blocks.

Solution.

Step 1: Introduce the uniform lower bound.

Let

$$\varepsilon := \inf_{i \in S} p_{i,i_0}^{(K)} > 0.$$

Then, no matter where the chain is at the start of a K -step block, the probability of hitting i_0 by the end of that block is at least ε .

Step 2: Bound the probability of avoiding i_0 for one block.

For any state i ,

$$\mathbb{P}_i(\tau_{i_0} > K) \leq 1 - \varepsilon.$$

Step 3: Iterate over successive blocks.

By the Markov property, after each block of K steps, the same bound applies again. Hence

$$\mathbb{P}_i(\tau_{i_0} > mK) \leq (1 - \varepsilon)^m, \quad m = 1, 2, \dots$$

Step 4: Convert the block bound into an exponential bound in n .

For arbitrary n , let $m = \lfloor n/K \rfloor$. Then

$$\mathbb{P}_i(\tau_{i_0} > n) \leq \mathbb{P}_i(\tau_{i_0} > mK) \leq (1 - \varepsilon)^m.$$

Since

$$(1 - \varepsilon)^m \leq \exp(-m\varepsilon) \leq \exp\left(-\frac{\varepsilon}{K}n + \frac{\varepsilon}{K}\right),$$

we may take

$$C_1 = e^{\varepsilon/K}, \quad C_2 = \varepsilon/K.$$

Any positive constants with the same exponential rate also work.

Final answer.

$\mathbb{P}(\tau_{i_0} > n \mid X_0 = i) \leq C_1 e^{-C_2 n} \text{ for suitable } C_1, C_2 > 0.$
